

ENHANCED ANTI-STOKES RAMAN SCATTERING FROM LIVING CELLS OF *CHLORELLA PYRENOIDOSA*

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The anti-Stokes/Stokes ratios for living *Chlorella pyrenoidosa* cells at 294 K considerably exceed those given by the thermal population factors $[n/(n+1)]$. We have shown that resonance enhancement in the scattered photons is mainly responsible for this, but cannot rule out that some population enhancement occurs.

Recent studies [1,2] have shown that the investigation of Raman scattering from photosynthetic systems can provide information on the chemical and structural composition of the absorbing pigments, and the molecular surroundings in which they are found. Resonances of the incident laser light with visible absorption bands of these pigments lead to an enhancement of the scattering. The use of such resonances to investigate specific molecules in heterogeneous biological systems is rather widespread [3].

We have measured Raman spectra of active *Chlorella pyrenoidosa* cells at 294 K in an attempt to clarify questions related to the role of vibrational states in biological membranes as predicted by theoretical considerations [4-7] and as suggested by some experimental observations [8,9]. According to the theory, a non-thermal population of vibrational states can occur in living systems which should lead to enhanced anti-Stokes scattering.

We have observed enhanced anti-Stokes scattering for several modes in *Chlorella pyrenoidosa* but point out that a large part of this arises from resonances in the scattered photon channel which is different for Stokes and anti-Stokes photons. Population factors derived from our experimental data show, however, that we cannot rule out that some population enhancement occurs.

The starting culture of *Chlorella pyrenoidosa* cells was obtained from D. Fork, Carnegie Institution, Stanford, and grown in nutrient solution as proposed by Lorenzen [10]. A CO₂-air mixture (1.5% CO₂) was bubbled through the cell culture which was illuminated with fluorescent lamps. The Raman measurements were performed with a flow arrangement as described elsewhere [11] to avoid heating the cells with the laser beam and to be able to keep them in a rather adapted state. The cell concentration was 10⁷ cells/ml with a flow speed of 40 cm/s. White background light was available for illumination of the reservoir in order to investigate a possible relation between Raman intensities and the activity of photosynthesis. All spectra were recorded at ~5 cm⁻¹ resolution using Ar⁺-laser excitation, a 1 m-double monochromator and conventional photon counting detection.

The Stokes Raman spectrum of active *Chlorella pyrenoidosa* cells at 294 K is shown in fig. 1 for a laser wavelength of 5145 Å (19 436 cm⁻¹). The main features are a few groups of sharp lines (width ~10 cm⁻¹) between 800 and 1600 cm⁻¹ superimposed on a rather structureless background. The anti-Stokes Raman spectrum for this range is shown in fig. 2 for the same laser frequency and power.

The intensity ratios R of the main anti-Stokes lines with respect to their Stokes counterparts ($R = I_{as}^{19436}/I_s^{19436}$) deviate considerably from the thermal population factor $[n(\Omega, T)/(n(\Omega, T) + 1)]$ which is

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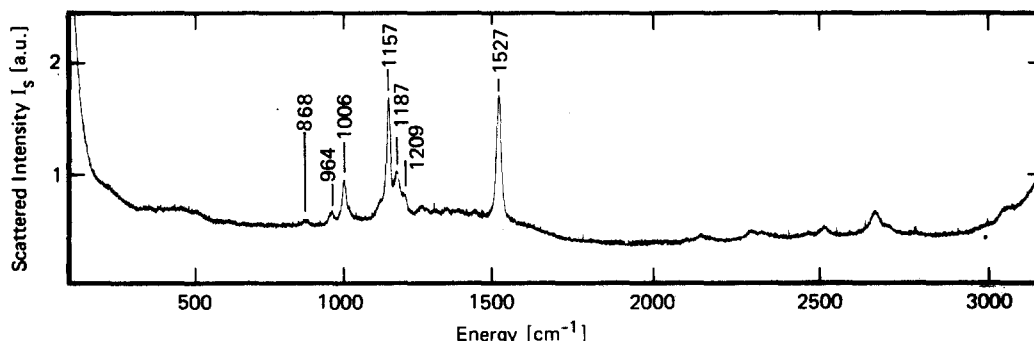


Fig. 1. Stokes Raman spectrum measured at 294 K with active cells of the green alga *Chlorella pyrenoidosa* during irradiation at 5145 Å. The bands at 868 cm⁻¹, 964 cm⁻¹, 1006 cm⁻¹, 1157 cm⁻¹, 1187 cm⁻¹, 1209 cm⁻¹ and 1527 cm⁻¹ are assigned to vibrational modes of carotenoid molecules [1].

given by a Planck equilibrium (Ω is the vibrational mode frequency, $T = 294$ K, see table 1, column 3). The dependence of the anti-Stokes intensity on laser power shows that local heating by the laser is not responsible for its enhancement.

As the laser wavelength was changed between 5145 Å and 4545 Å substantial changes in the scattered Stokes intensities occurred which overwhelmed the non-resonant ω^4 behaviour (fig. 3), and are attributed resonance with the absorption band of carotenoid pigments.

Furthermore, the influence of additional light (i.e., illumination of the cells in the reservoir with white light) on the intensity of the Stokes lines at 868 cm⁻¹, 1006 cm⁻¹ and 1527 cm⁻¹ was measured. Our preliminary results show that a reproducible increase in the intensity of those lines occurred which amounted to ~5% and which reached its maximum value approximately 60 s after light was switched on.

We concentrate our attention on the enhanced anti-Stokes/Stokes ratios R of the Raman lines at 868 cm⁻¹, 1006 cm⁻¹, 1157 cm⁻¹ and 1527 cm⁻¹. These are assigned to vibrations of carotenoid molecules by Lutz [1].

The intensity ratio R for excitation at laser frequency ω is given by eq. (1):

$$R(\Omega, \omega, T) = \frac{I_{as}^\omega}{I_s^\omega} = a \frac{(x - \Omega)^2}{(x + \Omega)^2} P(\Omega, T), \quad (1)$$

where a accounts for reabsorption of Stokes and anti-Stokes light, the second factor with $x = \omega - \omega_e$ expresses the resonance term, and P is the population factor which in normal thermal equilibrium is just $n/(n+1)$. Relaxing the assumption of a single resonance does not affect the essentials of our arguments.

Far from resonance both a and Ω can be neglected and R measures the population factor. On the other hand near resonance, and particularly for large Ω ,

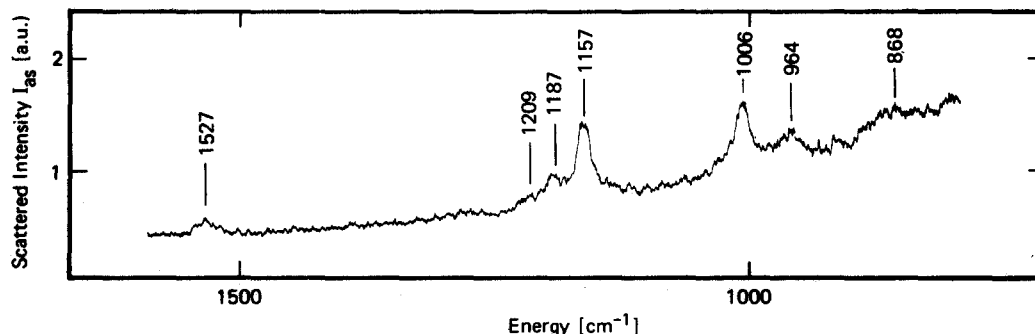


Fig. 2. Anti-Stokes spectrum between 800 and 1600 cm⁻¹ measured under the same experimental conditions as the Stokes spectrum shown in fig. 1. Each Stokes shifted line has its counterpart in the anti-Stokes spectrum. The intensity ratio $I_{as}^{19436}/I_s^{19436}$ appears to be much larger than given by thermal population factors assuming a Planck equilibrium.

Table 1

Vibrational energy Ω (cm ⁻¹)	Observed ratio R $I_{as}^{19436}/I_s^{19436}$	Anti-Stokes enhancement $R/n(n+1)$	Res. factor $I_s^{19436+\Omega}/I_s^{19436}$ from fig. 3	$\frac{P}{n/(n+1)}$
868	0.1	10.3	4.0	2.6±0.5
1006	0.060	7.2	5.6	1.3±0.1
1157	0.020	5.3	4.1	1.3±0.1
1527	0.0024	3.5	2.7	1.3±0.2

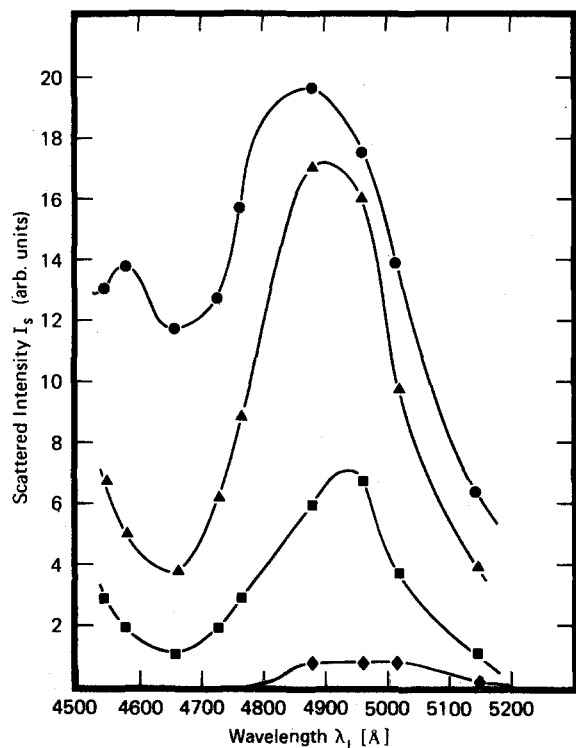


Fig. 3. Intensity dependence of the Stokes shifted bands at 868 cm⁻¹ (●), 1006 cm⁻¹ (■), 1157 cm⁻¹ (▲) and 1527 cm⁻¹ (○) on the wavelength of the incident laser light.

R can be dominated by the frequency-dependent cross section factor. We have attempted to separate this from the population factor P by considering the ratio of Stokes intensity $I_s^{\omega+\Omega}$ (laser at $\omega + \Omega$) and anti-Stokes intensity I_{as}^{ω} (laser at ω). In this configuration it can be shown from the usual equations for Raman intensities that

$$P = I_{as}^{\omega}/I_s^{\omega+\Omega} = R(I_s^{\omega}/I_s^{\omega+\Omega}), \quad (2)$$

is independent of resonance and reabsorption factors.

Eq. (2) can be rewritten to show the anti-Stokes enhancement as:

$$\frac{R}{n(n+1)} = \frac{I_s^{\omega+\Omega}}{I_s^{\omega}} \frac{P}{n(n+1)} \quad (3)$$

Table 1 shows values for the factors in eq. (3), the ratio $I_s^{\omega+\Omega}/I_s^{\omega}$ being obtained from fig. 3.

The population ratio P appears to be enhanced by factors ranging from 1.3 to 2.6 if compared with the thermal population factor $n/(n+1)$ (enhanced population factors have been suggested by refs. [4–7]).

However, the major part of the experimentally measured anti-Stokes enhancement can be explained as due to a resonance with the carotenoid absorption band at ~480 nm. This shows clearly the importance of an analysis of resonance enhancements for investigations of population effects.

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